

Multiple Choice (10 marks)

1. The anhydride of $\text{Ba}(\text{OH})_2$ is
 - (a) BaH_2
 - (b) BaOH
 - (c) Ba
 - (d) BaO
2. The +1 oxidation state is more stable than the +3 oxidation state for which group 13 element?
 - (a) B
 - (b) Al
 - (c) In
 - (d) Tl
3. Reduction of D-xylose with NaBH_4 yields a product that is a
 - (a) racemic mixture
 - (b) single pure enantiomer
 - (c) mixture of two diastereomers in equal amounts
 - (d) meso compound
4. The T_2 of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.
 - (a) 0.0471 Hz
 - (b) 21.2 Hz
 - (c) 42.4 Hz
 - (d) 66.7 Hz
5. The ^{13}C spectrum of which isomer of C_6H_{14} has lines with three distinct chemical shifts?
 - (a) hexane
 - (b) 2-methylpentane
 - (c) 3-methylpentane
 - (d) 2,3-dimethylbutane

True / False (10 marks)

Answer True or False

6. True or False: The ^{55}Mn nuclide has three allowed energy levels for its nuclear states.
7. True or False: The equilibrium polarization of ^{13}C nuclei in a 20.0 T magnetic field at 300 K is 1.71×10^{-5} .
7. True or False: Arsenic-doped silicon is classified as an n-type semiconductor due to the addition of donor impurities.
8. True or False: The infrared absorption frequency shift between DCl and HCl is primarily caused by differences in their dipole moments.
9. True or False: Silicon is most commonly found in nature as an oxide, primarily in the form of silicates and quartz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the reasons why spin trapping is a preferred method for detecting free radical intermediates in chemical reactions, focusing on the advantages of detecting spin adducts.
12. Discuss the rate expression for a zero-order chemical reaction, specifically how it relates to the rate constant, and analyze its implications for reaction kinetics.

End of Paper